

Logarithmic Functions + DOK-3 Sales Revenue Inverse (Item 28)

Lesson 6-4 · Quick · Teacher Edition

OBJECTIVES

Math Objective	Graph logarithmic functions, identify key features (asymptote, domain, range, intercepts), and find equations of inverses of exponential and logarithmic functions.
Language Objective	Describe how transforming a log function shifts its asymptote and key points; explain why the inverse of a log is an exponential.
Essential Understanding	Log and exponential functions are inverses — swap x and y , solve. The asymptote of a log is the y -axis mirror of an exp function's horizontal asymptote.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Fluency with graphing log functions and finding inverses; apply to the Sales Revenue DOK-3 task (Item 28 rehearsal for Form B model-then-invert structure).
Essential Question	How do the graphs of logarithmic and exponential functions relate, and how do we find one from the other?
Materials	Student packet, pencil, calculator. Rules card: $\log_b 1 = 0$; $\log_b b = 1$; $y = \log_b x$ has VA at $x = 0$; $y = \log_b(x - h) + k$ has VA at $x = h$; inverse of $y = a \cdot b^x$ is $y = \log_b(x/a)$; inverse of $y = \log_b(x - h) + k$ is $y = b^{(x-k)} + h$.

[PIN] PRE-FLIGHT — SINGLE-PERIOD COMPRESSED LESSON + DOK-3 CAPSTONE

This is a single-period quick treatment of Lesson 6-4. ONE example modeled in Launch (Ex 3B: inverse of $g(x) = \log_7(x + 5) +$ composition verification). Contrast with Ex 3A (inverse of exponential) in 90 sec only — same swap-and-solve process, direction reversed. Release to Explore at minute 15.

DOK-3 driver: Practice #28 ($R = 12 \log(a + 1) + 25$). Students (a) find the inverse function $a = 10^{\frac{R-25}{12}} - 1$, (b) judge which form is easier for specific R values. Real Savvas values from SE source.

Pace: [STAR] q28 (10 min) → q21 (4 min) → q24 (4 min) → q13 (4 min) → q9 (3 min). ~25 min total Explore.

45-min Wed F variant: cut q9. Never cut q28.

Assessment alignment: q21→Form B Q7, q24→Q12, q13→Q11, q9→Q9/Q10, q28→DOK-3 rehearsal.

Tag: bridge to log inverses launch

Do Now

DOK: 1–2

Minutes: 7

Teacher does

- Hand out at door.
- Circulate 4 min.
- Cold-call on q31: ask which transformation shifts the VA.
- Pull sheets at minute 7.

Students do

- q31: translation multi-select for $h(x) = \ln(x + 2) - 1$.
- q5: graph $y = \log_4 x$ with all key features.
- q32: find inverse of $f(x) = 5^{x+1}$ (SAT/ACT).

Questions to ask

- q31: how does $(x + 2)$ inside the log change the VA?
- q5: domain? VA? x-intercept? End behavior as $x \rightarrow 0^+$ and $x \rightarrow \infty$?
- q32: undo 5^x using log base 5 — what is the inverse operation?

Adult role

Solo teacher: circulate silently. No hints on q32.

Do Now 1 — Translation identification: $h(x) = \ln(x + 2) - 1$

The logarithmic function $g(x) = \ln x$ is transformed to $h(x) = \ln(x + 2) - 1$. Which of the following are true? Select all that apply.

- ☐ $g(x)$ is translated 2 units upward.
- ☐ $g(x)$ is translated 2 units to the right.
- ☐ $g(x)$ is translated 2 units to the left.
- ☐ $g(x)$ is translated 1 unit downward.
- ☐ $g(x)$ is translated 1 unit to the left.
- ☐ The vertical asymptote shifts 2 units to the left.
- ☐ The vertical asymptote shifts 2 units to the right.

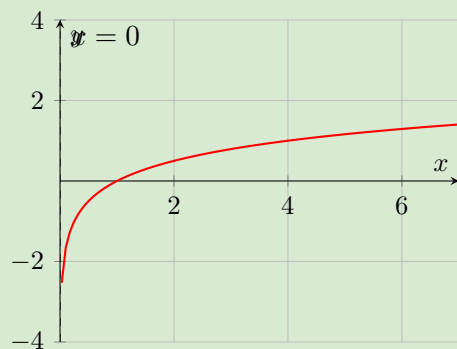
[CHECK] ANSWER KEY

Answer key (from source): $g(x)$ is translated 2 units to the left; $g(x)$ is translated 1 unit downward; The vertical asymptote shifts 2 units to the left.

Do Now 2 — Graph $y = \log_4 x$; state key features

Graph the function $y = \log_4 x$ and identify the domain and range. List any intercepts or asymptotes. Describe the end behavior.

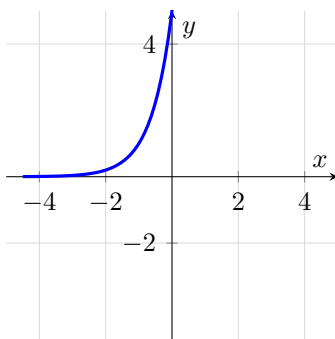
[CHECK] ANSWER KEY



Answer key (from source): Domain: $\{x \mid x > 0\}$; range: all real numbers; x -intercept: 1; asymptote: y -axis; end behavior: As $x \rightarrow 0$, $y \rightarrow -\infty$. As $x \rightarrow \infty$, $y \rightarrow \infty$.

Do Now 3 — SAT/ACT: inverse of $f(x) = 5^{x+1}$

SAT/ACT The graph shows the exponential function $f(x) = 5^{x+1}$. Which of the following functions represents its inverse, $f^{-1}(x)$?



- Ⓐ $f^{-1}(x) = 1 + \log_5 x$ Ⓑ $f^{-1}(x) = \log_5 x - 1$
 Ⓒ $f^{-1}(x) = \log_5(x - 1)$ Ⓓ $f^{-1}(x) = \log_5(x + 1)$

[CHECK] ANSWER KEY

Answer key: Ⓑ $f^{-1}(x) = \log_5 x - 1$. Swap x and y : $x = 5^{y+1}$, so $y + 1 = \log_5 x$ and $y = \log_5 x - 1$.

[CHECK] ANSWER KEY

Answer key (from source): $g(x)$ is translated 2 units to the left; $g(x)$ is translated 1 unit downward;
 The vertical asymptote shifts 2 units to the left.

(verify before class)

(verify before class)

Tag: teacher models Ex 3B (inverse of log) + 90s Ex 3A contrast

Launch

DOK: 2

Minutes: 8

Teacher does

- Project Ex 3B. Think aloud: $g(x) = \log_7(x + 5)$. Step 1: swap x and y . Step 2: write $7^x = y + 5$. Step 3: solve — $y = 7^x - 5$.
- “Now verify with composition: $g(g^{-1}(x)) = \log_7((7^x - 5) + 5) = \log_7(7^x) = x$. ✓”
- 90-sec contrast: Ex 3A (inverse of exponential, e.g. $y = 3 \cdot 5^x$). “Swap x and $y \rightarrow x = 3 \cdot 5^y \rightarrow x/3 = 5^y \rightarrow y = \log_5(x/3)$. Same swap-and-solve.”
- Flag the Rules card — students copy VA shift rule and inverse rules into margin.
- 30-sec pause: partner repeats the inverse steps in their own words.
- Do NOT solve Explore items — release at minute 15.

Students do

- Watch Ex 3B silently. Note swap-then-solve sequence.
- Copy composition verification step.
- During 90-sec contrast: note that inverting an exponential gives a log, and inverting a log gives an exponential.
- Copy Rules card VA and inverse rules into margin.

Questions to ask

- In $g(x) = \log_7(x + 5)$, what is the domain? VA? ($x = -5$)
- After swapping x and y , how do we isolate y when $7^x = y + 5$?
- What is $g^{-1}(x)$? What is its domain? ($x \in \mathbb{R}$)
- Why does the VA of g become the HA of g^{-1} ?

Adult role

Project Ex 3B. Model composition check. 90-sec Ex 3A contrast. Release promptly at minute 15.

Example 3B

Inverses of Exponential and Logarithmic Functions

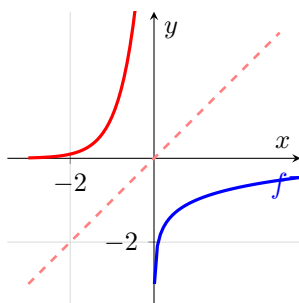
What is the equation of the inverse of the functions?

A. $f(x) = 10^{x+1}$

$$\begin{aligned}
 y &= 10^{x+1} \\
 x &= 10^{y+1} \quad (\text{Rewrite the function as } y =, \text{ interchange } x \text{ and } y, \text{ and solve for } y.) \\
 y + 1 &= \log x \\
 y &= \log x - 1
 \end{aligned}$$

The equation of the inverse of $f(x) = 10^{x+1}$ is $f^{-1}(x) = \log x - 1$.
Verify by function composition.

$$\begin{aligned}
 f(f^{-1}(x)) &= 10^{(\log x - 1) + 1} & f^{-1}(f(x)) &= \log 10^{x+1} - 1 \\
 &= 10^{\log x} & &= x + 1 - 1 \\
 &= x & &= x
 \end{aligned}$$



CONCEPTUAL UNDERSTANDING / LOOK FOR RELATIONSHIPS

$f(x)$ is a translation of the parent function $y = 10^x$ one unit left. $f^{-1}(x)$ is a translation of the parent function $y = \log x$ one unit down. Graphical translations of a function and its inverse are directly related. Explain.

B. $g(x) = \log_7(x + 5)$

$$y = \log_7(x + 5) \quad (\text{Rewrite as } y =.)$$

$$x = \log_7(y + 5)$$

$$y + 5 = 7^x$$

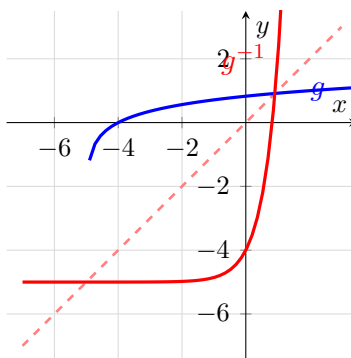
$$y = 7^x - 5$$

The equation of the inverse of $g(x) = \log_7(x + 5)$ is $g^{-1}(x) = 7^x - 5$.

Verify by function composition.

$$\begin{aligned} g(g^{-1}(x)) &= \log_7(7^x - 5 + 5) \\ &= \log_7 7^x \\ &= x \end{aligned}$$

$$\begin{aligned} g^{-1}(g(x)) &= 7^{\log_7(x+5)} - 5 \\ &= x + 5 - 5 \\ &= x \end{aligned}$$



[CHECK] ANSWER KEY

The equation of the inverse of $f(x) = 10^{x+1}$ is $f^{-1}(x) = \log x - 1$.

The equation of the inverse of $g(x) = \log_7(x + 5)$ is $g^{-1}(x) = 7^x - 5$.

[MIC] TEACHER SCRIPT

1. "Watch me find the inverse of a LOG function."
2. "Step 1: replace $g(x)$ with y . Step 2: swap x and y ."

3. “Step 3: rewrite in exponential form — $7^x = y + 5$.”
4. “Step 4: solve for y — $y = 7^x - 5$. That’s $g^{-1}(x)$.”
5. “Composition check: $g(g^{-1}(x)) = \log_7(7^x - 5 + 5) = x$. ✓”
6. “Now 90 seconds: Ex 3A — inverting an exponential gives a log. Same swap-and-solve.”
7. Release to Explore.

Tag: TPS + DOK-3 driver

Explore

DOK: 2–3

Minutes: 25

Teacher does

- 4 laps across 25 min.
- Lap 1 (10 min): [STAR] q28 DOK-3. Press pairs on inverse setup and form comparison. DO NOT give the inverse formula — redirect with ‘what does it mean to undo R ?’
- Lap 2 (4 min): q21. Press notation: what is the inverse of 5^{x-3} ?
- Lap 3 (4 min): q24. Press: inverse of $\log_2(8x)$ — watch for domain.
- Lap 4 (4 min): q13. Press: how does the horizontal shift move the VA?
- Lap 5 (3 min): q9. Are graphs inverses? Key test: reflective symmetry over $y = x$.
- 45-min Wed F: skip q9.

Students do

- 5 items in order: [STAR] q28 → q21 → q24 → q13 → q9.
- For q28: BEFORE computing, write down $R = 12 \log(a + 1) + 25$ and plan how to isolate a .
- For q28 part 2: decide which form (original or inverse) is easier for $R = 49$ and $R = 61$. Write a justification sentence.

Questions to ask

- q28: how do you undo $+25$ first? Then undo $\times 12$?
- q28: after $(R - 25)/12 = \log(a + 1)$, what is the next step?
- q28: for $R = 49$, would you use $R = 12 \log(a + 1) + 25$ or $a = 10^{\frac{R-25}{12}} - 1$? Why?
- q21: identify base and exponent — what is the log that undoes 5^x ?
- q24: inverse of $\log_2(8x)$ — swap x and y first. Then rewrite.
- q13: find the VA of the translated log from the graph.
- q9: if two graphs are inverses, what visual test do they pass?

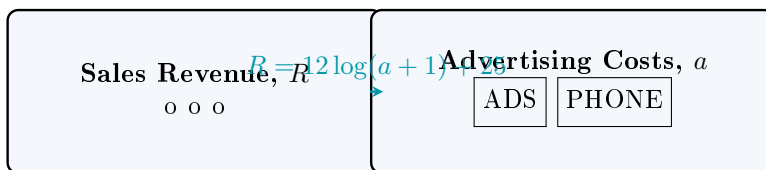
Adult role

Solo teacher: 4 laps. On q28, press the judgment step (which form and why) — that is the DOK-3 performance criterion.

Practice #28 [STAR] DOK-3 SALES REVENUE INVERSE — Form B model-then-invert rehearsal

A company uses this function to relate sales revenue, R , and advertising costs, a . What is the equation

of the inverse of the equation? Which equation would be easier to use to find a value of a for a particular value of R ?



[STAR] DOK-3 SALES REVENUE — Practice #28

Students must (a) isolate a from $R = 12 \log(a + 1) + 25$, yielding $a = 10^{\frac{R-25}{12}} - 1$, (b) judge which form is easier for specific R values.

Key criteria: (1) correct algebraic isolation steps shown, (2) inverse formula correct (base 10, not natural log), (3) justification for form preference cites a specific computation.

Assessment alignment: DOK-3 rehearsal for Form B model-then-invert structure

[CHECK] ANSWER / ANTICIPATED TALK

The inverse of the formula is $a = 10^{\frac{R-25}{12}} - 1$. It would be easier to use the inverse to find a when R is known.

Assessment alignment: DOK-3 rehearsal for Form B model-then-invert structure

Practice #21 — Inverse of exponential 5^{x-3} [Form B Q7 rehearsal]

Find the equation of the inverse of each function. $f(x) = 5^{x-3}$.

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): $f^{-1}(x) = \log_5 x + 3$

Assessment alignment: Form B Q7: inverse of $5^{(x-3)}$

Practice #24 — Inverse of $\log_2(8x)$, token drag [Form B Q12 rehearsal]

Find the equation of the inverse of each function. $f(x) = \log_2(8x)$.

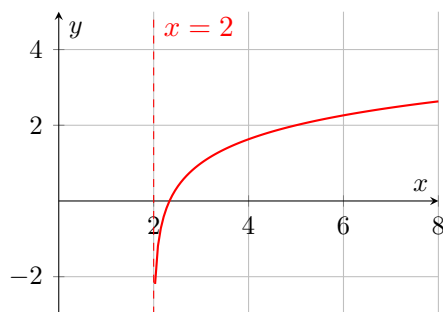
[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): $f^{-1}(x) = \frac{2^x}{8}$

Assessment alignment: Form B Q12: inverse of $\log_2(8x)$, token drag

Practice #13 — Asymptote of translated log from graph [Form B Q11 rehearsal]

Use Structure The graph shows a transformation of the parent graph $f(x) = \log_3 x$. Write an equation for the graph.



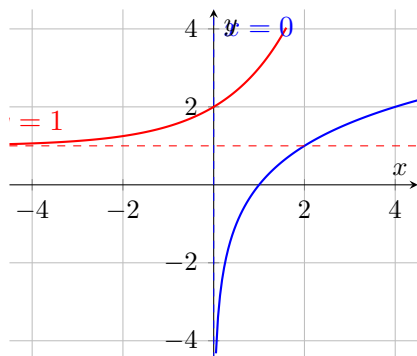
[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): $g(x) = \log_3(x - 2) + 1$

Assessment alignment: Form B Q11: asymptote of translated log from graph

Practice #9 — Are graphs inverses? Inverse from graph [Form B Q9/Q10 rehearsal]

Look for Relationships Are the logarithmic and exponential functions shown inverses of each other? Explain.



[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from source): No; The x -intercept of the blue function is not equal to the y -intercept of the red function.

Assessment alignment: Form B Q9/Q10: are graphs inverses; inverse from graph

Tag: academic conversation + 6-5 bridge

Share / Summary

DOK: 2

Minutes: 5

Teacher does

- Cold-call 1 pair to present q28 solution and form-choice justification.
- Class signals thumbs. Write $a = 10^{\frac{R-25}{12}} - 1$ on board.
- 6-5 bridge: “Log properties (product/quotient/power rules) will let you expand and condense log expressions — making inverses and equations faster.”

Students do

- Presenting pair: read inverse formula and cite which form is easier for $R = 49$ and $R = 61$.
- Rest of class: signal thumbs and write bridge note in margin.

Questions to ask

- How did you isolate a from $R = 12 \log(a + 1) + 25$?
- For $R = 49$, which form is computationally easier? Why?

Adult role

Cold-call a pair that struggled on the judgment step — hold them accountable to citing a computation, not just a preference.

Tag: inverse + asymptote (not graded)

Exit Ticket**DOK: 1-2****Minutes: 5****Teacher does**

- Collect at door.
- Scan: (1) Did students swap x/y correctly? (2) Did students solve for y in exponential form? (3) Did they state VA of f as $x = 2$ and HA of f^{-1} as $y = 2$?

Students do

- Two-part custom exit: find inverse, then state both asymptotes.

Questions to ask

- Inverse of $f(x) = 3 \log_5(x - 2) + 1$: swap $\rightarrow$ exponential form $\rightarrow$ solve.
- VA of f is $x = \underline{\hspace{1cm}}$ (vertical asymptote of original log).
- HA of f^{-1} is $y = \underline{\hspace{1cm}}$ (the VA of f becomes the HA of the inverse).

Adult role

Collect. No grade. Use responses to plan any re-teach before 6-5.

[CLIPBOARD] EXIT TICKET — Student Form

Find the inverse of $f(x) = 3 \log_5(x - 2) + 1$.

State the vertical asymptote of f .

State the horizontal asymptote of f^{-1} .

One biggest thing I learned today: _____

ANSWER KEY (TE only): $f^{-1}(x) = 5^{\frac{x-1}{3}} + 2$. VA of f : $x = 2$. HA of f^{-1} : $y = 2$.

[CLOCK] PERIOD A / PERIOD F — TIMING NOTES

50-min base: standard pacing above.

45-min Wed F variant: drop q9 from Explore. Never cut q28.

If finished early: hand out Practice #33 (Telescope PT — stretch).

[PUZZLE] MODIFICATIONS / SUPPORT FOR IEP STUDENTS

- Provide the Log Function Key Features + Inverse Rules card as a sticker/cut-out.
- For q28, provide the setup: $(R - 25)/12 = \log(a + 1)$. Students solve from that point.
- Accept verbal justification for form-choice in lieu of written sentence.

[GLOBE] SUPPORTS FOR ELL STUDENTS

- Sentence frames on student page (don't skip).
- Word cards: 'vertical asymptote' = a line the graph never touches; 'inverse function' = reverses the input/output; 'composition' = plugging one function inside another.
- 'log base b of x ' = the exponent you put on b to get x .
- Pair ELL students with English-dominant partners for TPS.